Mohammed Khaled

Data Scientist and Computational Researcher

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Professional Summary

Data Scientist and Computational Researcher with 6+ years of experience developing machine learning models, computational tools, and scientific software for large-scale molecular simulation data. Experienced in Python, PyTorch, Scikit-learn, statistical modeling, feature engineering, and high-performance computing (HPC). Proven track record of transforming terabyte-scale datasets into actionable insights, building reproducible ML workflows, and collaborating with multidisciplinary international teams. Passionate about applying AI and data science to solve complex real-world problems.

Work Experience

Postdoctoral Researcher — Aug. 2025 – Present
Johannes Gutenberg University Mainz — Mainz, Germany

- Analyzed high-dimensional biomolecular dynamics using Python-based machine learning and molecular dynamics simulations, resulting in two peer-reviewed publications
- Applied machine learning and statistical analysis to identify hidden structural patterns and molecular interaction mechanisms

Postdoctoral Researcher — May 2024 – July 2025
University of Applied Sciences Bonn-Rhein-Sieg — Bonn, Germany

- Implemented GeoMol++, a Python/PyTorch generative deep learning pipeline extending the GeoMol architecture with non-bonded edges in graph neural networks for molecular conformation generation, developing end-to-end preprocessing, training, and evaluation pipelines that achieved meaningful RMSD improvements over baseline

Postdoctoral Researcher — June 2023 – Oct. 2023
Forschungszentrum Jülich — Jülich, Germany

- Transformed raw simulation outputs into interpretable structural models in collaboration with international research teams

Ph.D. Researcher — Dec. 2019 – May 2023
Forschungszentrum Jülich — Jülich, Germany

- Developed ATRANET, a Python-based framework for automated construction of transition networks for biomolecular systems
- Built scalable data pipelines processing terabyte-scale atomistic simulation datasets using Python and HPC environments
- Administered Linux-based HPC infrastructure, supporting computational workloads for multiple researchers and optimizing resource utilization
- Applied unsupervised machine learning (K-means clustering, PCA) and Markov state models to discover metastable states in high-dimensional time-series data, enabling quantitative analysis of complex molecular dynamics

Education

Ph.D. in Mathematics and Natural Sciences — 2019 – 2023
Heinrich Heine University Düsseldorf — Düsseldorf, Germany

- Thesis: Network Models and Structural Characteristics of Oligomer Formation — studied protein aggregation mechanisms using large-scale molecular dynamics simulations and network-based modeling approaches
- Focus: computational modeling of complex dynamical systems, Markov state models, statistical physics and high-dimensional data analysis

M.Sc. in Physics — 2016 – 2019
Birzeit University — Palestine

- Thesis: Comparative analysis of the conformational dynamics of Ras mutants using molecular dynamics simulations and Markov state models
- Focus: statistical mechanics, computational physics, and time-series analysis of molecular systems
- Summer School Internship. Institute of Applied Physics, University of Münster, Münster, Germany, Jun. 2017 – Aug. 2017

B.Sc. in Physics 2012 – 2016
Birzeit University Palestine

Skills

Programming: Python, Bash, MATLAB, C++, SQL
Python Ecosystem: NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, SciPy, TensorFlow
Machine Learning: Deep Learning, Dimensionality Reduction, Clustering, Generative AI, Feature Engineering, Hyperparameter Optimization
Scientific Computing: High-Performance Computing (HPC), SLURM, Parallel Computing, Linux, Data Visualization, Docker
Software Development: Object-Oriented Programming (OOP), Git/GitHub, Data Processing Pipelines

Professional Development

Linux System Administration 2025
Com Training and Services Mainz/Wiesbaden

High-Performance Computing with Jupyter 2023
Jülich Supercomputing Centre

Introduction to Machine Learning 2022
Helmholtz AI

Python for Data Science and Machine Learning Bootcamp 2021
Udemy

Core Competencies

Analytical Thinking, Problem Solving, Cross-functional Collaboration, Scientific Communication, Adaptability, Continuous Learning, Critical Thinking, Time Management, Project Management, Mentoring

Languages

English (Fluent), German (Basic)

Publications

9 peer-reviewed publications in international journals. Full list: Google Scholar Profile